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Forming Connected Multi-Robot Teams Utilizing Aerial Robots

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***Abstract*—In this paper, we study the connected multi-robot team formation problem. We consider the settings where multiple aerial and ground robots with limited communication ranges are operating in an unbounded environment. The ground robots do not have a priori knowledge about the initial locations of each other and the aerial robots. Moreover, the aerial robots do not know the initial locations of the ground robots in advance. Our goal is to divide the robots into teams as quickly as possible by relocating them. Each team should be of same size and include one aerial robot assigned to it as its leader, and the final configurations of the robots in each team should form a fully connected network topology. To solve this problem, we present an algorithm that combines distributed coordination and online planning of the robots. We evaluate the performance of the proposed algorithm in simulations.**

***Index Terms*—Multi-Robot Systems, Distributed Robot Systems, Multi-Robot Teams, Team Formation.**

I. Introduction

Multi-robot systems (MRS) can perform various challenging tasks through exchanging information, making decisions collaboratively, and taking advantage of heterogeneous capabilities (e.g. sensing, recharging, object manipulation, communication, and computation). Their ability to perform complex tasks efficiently along with other significant advantages such as scalability, robustness, and cost-effectiveness makes the utilization of MRS in various applications more attractive than a single sophisticated robot. For instance, MRS can provide solutions for versatile applications including search and rescue [1], agriculture [2], exploration [3], construction [4], package delivery [5], target trapping [6], perimeter defense [7], and whale rendezvous [8].

Robots typically achieve collaboration through communication. However, the diversity of the tasks assigned to the robots with limited communication ranges can cause them to work in remote locations across a large area. Moreover, the network can be disconnected, the robots can be outside the communication range of each other and unaware of the locations of each other. Consider that the next mission that arise requires the robots to work collaboratively, thus the robot should come together with the other robots. Motivated by this scenario, in this paper, we study the problem of forming aerial-ground robot teams as quickly as possible. The goal is to relocate the aerial and ground robots such that each team is of same size and has one aerial robot as its leader (team coordinator), and the final configurations of the robots in each team form a fully connected network topology.

Our problem is related to the researches including the rendezvous, network formation, and coalition formation. The goal of the rendezvous problem is the robots to meet at a location that is not predetermined as quickly as possible [9]–[11]. The first version of this problem interests in designing search strategies for the rendezvous of all robots [10], [11] or only a group of robots [9]. The proposed online strategies focus on solving the problem without requiring the robots to know the locations of each other in advance. Another version of the problem is studied in the field of distributed cooperative control of multi-agents [12]–[14]. This version interests in the robot's mobile target tracking also its navigation towards the target. In contrast to the first version, the solutions proposed for this version include the robot observing the states of its neighbors (e.g. position, orientation, velocity information) using its (long range) sensory system. Parasuraman *et al.* [12] studied the problem of gathering robots in different groups rather than at a single consensus point, given that the robots initially know their neighbors. Considering that the robots are connected by a communication network, Hu *et al.* [14] developed a cluster formation containment coordination framework for networked robots that consist of the leaders and followers.

The objective in the network formation problem is to form a single connected network between the robots [15]–[23]. This problem is studied for both cases where the robot positions are known [18], [19] and not known [15], [17] in advance. All robots or only a single robot can be initially active [15], [16]. When there is initially one active robot while the rest of the robots are asleep [15], [16], the problem is also known as the Freeze-Tag Problem. The goal in this problem is to wake up all the asleep robots (e.g. to disseminate information) as quickly as possible. As the asleep robots are found and become active, they can also join the searching and information dissemination process [15]. Poduri and Sukhatme [17] studied the network formation problem for the robots at unknown locations using a decentralized approach for a bounded environment. They used a randomized motion strategy and showed that the connectivity time is $O(1/\sqrt{N})$ and $\Omega((\log N)/N)$, where N denotes the number of robots. For k-connected network formation problem, Engin and Isler [19] presented an algorithm that is $O(D)$-competitive given that the initial locations of the arbitrarily deployed robots are known in advance. D denotes the diameter of the smallest k-connected graph induced on the initial configuration of the robots. A graph is k-vertex-connected if it has more than k vertices and removing any combination of $k-1$ vertices does not make it disconnected.

Redeploying mechanisms for multi-robots were proposed also to maintain the connectivity between the group of robots moving to dynamic task areas [20] and between moving clients and a fixed base station [23]. A probabilistic connectivity-restoration strategy for a swarm robotic system that consists of a group of follower robots is presented in [21]. A leader robot is utilized to restore the swarm connectivity by searching for the lost robots in regions of interest. Lost robots are defined as the robots that are not connected to the leader robot. Instead of maintaining a connected network at all times, Hollinger and Singh [22] proposed an online algorithm to achieve periodic connectivity, where the network must regain connectivity at a fixed interval.

The diversity of the tasks require multiple robots with different and complementary capabilities to cooperate, e.g. through forming coalitions [24]–[27]. A coalition is referred to a team of robots that are collaboratively performing a task. The focus in coalition formation is task allocation among the robots rather than autonomous relocation of the robots to form teams as in our study. Dutta and Asaithambi [26] modeled the problem of multi-robot coalition formation for task allocation as a variant of bipartite matching. A coalition structure is defined as non-overlapping clusters that covers all the robots. To solve this problem, a heuristic algorithm was proposed.

The contributions of this work are as follows. For an autonomous multi-robot system with n ground robots and m aerial robots in an unbounded environment, we study the *multi-robot connected team formation problem* (CTFP). The goal in CTFP is to divide the robots into teams as quickly as possible by relocating them, such that (1) each team includes the same number of ground robots and exactly one aerial robot as the team leader and (2) the final configurations of the robots in each team form a fully connected network topology. To solve this problem, we propose an algorithm that combines the distributed coordination and the online planning of the robots. The solution does not require the aerial robots to know the locations of the ground robots and the ground robots to know the locations of each other and the aerial robots in advance. Moreover, unlike the aforementioned studies, we do not consider that the ground robots know their neighbors or are initially connected by a communication network. The performance of the algorithm is evaluated through simulations.

II. Problem Formulation

Consider a multi-robot system that includes n Aerial Robots (ARs) and m Ground Robots (GRs). The GRs are distributed uniformly at random in a circular area A of radius L. Multi-robot connected team formation process is initiated by the ARs. When this task needs to be executed, the ARs are deployed at a common location X_u within A. The GRs do not know the initial locations of each other and the ARs, while the ARs do not know the initial locations of the GRs. Moreover, no GR and AR know the distance and direction to any other GR. The GRs and ARs each has a unique integer identifier (*id*). The set of initial positions of the GRs is denoted by $X = \{x_1, \cdots, x_n\}$. Two agents (GR/AR) can communicate with each other if the distance between them is within R for a given fixed communication radius R. The following assumptions are made: (1) The GRs and the ARs know their own locations and move at unit speed, (2) $m|n$ (i.e. n/m is an integer), and (3) The ARs fly at a fixed altitude h, and know n, m, and L in advance.

III. Algorithm

In this section, we present our algorithm *Connected Team Builder* (CTB) to solve CTFP. The algorithm includes 5 main stages: Stage 1-5. In sections III-A-III-F, we explain these stages and present an example execution of CTB. The pseudocodes of CTB (starting from Stage-2) that are executed by the AR and the GR are presented in Algorithms 1 and 2, respectively.

A. Stage-1

In stage-1 of CTB, the ARs collaboratively explore the environment to find the GRs at unknown locations. Each AR explores the environment following concentric circles that are centered at X_u. The exploration proceeds in rounds indexed by $k \geq 1$. Each round consists of at most m phases. The total number of rounds executed by the ARs is $r^* \leq \lceil L/(2md) \rceil$, where $d = \lfloor \sqrt{R^2 - h^2} \rfloor$. Given that $r^* = \lceil L/(2md) \rceil$, the number phases that is executed in the last round is $p^* = \mathrm{mod}\ (\lceil L/(2d) \rceil, m)$ which also shows the number of ARs required to execute the last round.

The phases to be executed in round k are assigned to the ARs in decreasing order with respect to their distance traveled costs. Thus, the circle with the smallest radius (the innermost circle) is assigned to the AR that has the highest distance traveled cost until the end of the round $k-1$. The AR that is assigned to phase p is denoted by u, where $1 \leq p \leq m$. The rendezvous location of the ARs at the end of the current round k is denoted by $\mathcal{X}_k$, where $\|\mathcal{X}_k - \mathcal{X}_u\| \leq 2dmk$. To explore the assigned phase p of round k, u moves $2dp$ distance forward starting from $\mathcal{X}_{k-1}$, then follows a circle with radius $2d(km+p)$, and finally moves to $\mathcal{X}_k$. At the end of each round, the ARs meet to exchange the information of the GRs that they found. Thus, they check if all GRs are found or not. When this event occurs, the ARs terminate the exploration. While following a circular path, the AR covers a ring of $2d$. This ensures that any waiting GR is found by a hovering AR during its exploration. The example execution of stage-1 with four ARs is shown in Fig. 1, where $r^* = 2$.

B. Stage-2

Once all GRs are found, Stage-2 of CTB starts. This stage includes the following main steps executed by the ARs: (1) computing the Minimum Spanning Tree (MST) on the initial positions of the GRs, and (2) computing the updated positions of the GRs as if the message relaying phase has been executed.

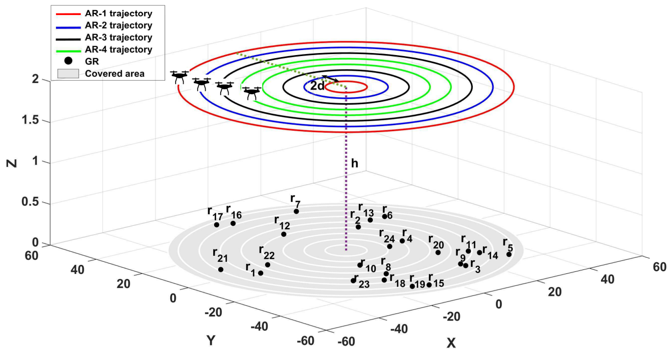

Fig. 1: Example execution of Stage-1 of CTB Algorithm. $n = 24$, $m = 4$, $R = 4$, $h = 2$, $L = 50$.

Let $G(V, E)$ be a complete graph on the initial positions of the GRs, where the vertices V are the initial locations of the GRs and the weights of edges E are the Euclidean distances between each pair of GRs. Prim's algorithm [28] is used to compute the MST $\mathcal{T}$ giving G and the root (i.e. the starting vertex of the tree) as inputs. The GR with the smallest id is chosen as $\mathcal{T}.root$.

Let msg denote the message that is created by the AR to be sent to a group of GRs. After $\mathcal{T}$ is computed, the goal of the ARs is to construct k teams with equal number of GRs and send msg to the leaf GRs in the MST. Starting from the leaf GRs in $\mathcal{T}$, the GRs then take over the information dissemination task and relay the received msg to each other. This step is called *message relaying* and aims to decrease the distance traveled by the ARs. During the execution of this step, the GR can change its position in order to relay the information to another GR. Therefore, the ARs determine the teams according to the positions of the GRs after the message relaying step which is denoted by set $X' = \{x'_1, \cdots, x'_n\}$. Since at the end of Stage-1, the ARs know X, they can compute X' in advance as follows.

msg consists of the following information: X', $\mathcal{T}$, and the constructed teams. In the message relaying step, when a GR receives msg, it checks whether it is the closest child to its parent or not. If it is, it moves toward its goal position in X'. The GR that sends msg to its parent is called the *relay GR*. Note that each GR that is not a leaf GR receives msg only from one of its children. The AR uses a geometrical transformation called homothety to determine the goal position that each GR should reach to communicate with its parent (refer to Algorithm 1, lines 4-16). Given a point $c \in \mathbb{R}^2$ called its center and a real positive number ϕ called its ratio, the homothety transforms a point $x \in \mathbb{R}^2$ with respect to c as

$$x' = c + \phi(x - c). \tag{1}$$

Considering x as the position of the relay GR and c as the position of the relay GR's parent in (1), we compute the goal position x'_1 of the relay GR located at x_1 as

$$x'_1 = c + \min(1, R/l)(x_1 - c), \tag{2}$$

where $l = \|c - x_1\|$ denotes the distance between the relay GR and its parent. Since $\phi = \min(1, R/l)$, the relay GR moves toward its parent only if $l > R$.

C. Stage-3

With X' computed, in Stage-3, we determine the centers and members of each team. For this data-partitioning step, we utilize the k-means++ algorithm [29] which improves the k-means algorithm [30] by using a randomizing seeding technique. Given the number of clusters k and a number of observations, k-means assigns the observations to exactly one of k clusters that are defined by their centroids (cluster centers). ARs execute the k-means++ algorithm with inputs k and observations that are set as m and X', respectively (refer to Algorithm 1, line 17). The set of teams formed by the k-means++ algorithm is denoted by $T = \{t_1, \cdots, t_m\}$ and the team of the GR with id a is denoted by $t(a)$. While the teams' centroid locations are obtained, the algorithm does not guarantee an equal number of GRs in each team. Therefore, we propose a method which relocates the points such that each cluster is of same size, a point is assigned to exactly one cluster, and the total standard error of the GR-team assignment is minimized. To solve this linear assignment problem in polynomial time, we use the Hungarian Algorithm (HA) [31] which takes a cost matrix $M = n \times n$ as input.

M is created as follows (refer to Algorithm 1, lines 18-32). We consider that the rows and columns of M represent the teams and the GRs, respectively. Since there are m teams, to create $n \times n$ matrix, the same row is inserted n/m times for each team. Thus, $M = ((n/m) \times m) \times n = n \times n$. Each value in M represents the standard error (SE) of the teams. To compute SE for a matrix cell $M_{i,j}$ at row i and column j, we temporarily remove GR-j from its team which was obtained by k-means++, and add it to team t_i. Then, we create an array D of size n which holds each GR's distance to its current team center. The standard deviation of the elements of D is denoted by $\sigma(D)$. Thus, $SE = \sigma(D)/n$. Given the defined M as input, HA returns the list of row indices that are assigned to the GRs (refer to Algorithm 1, line 33). The set of teams formed after executing HA is denoted by $T' = \{t'_1, \cdots, t'_m\}$. Since every n/m consecutive rows in M are the same, the returned row index i for GR-j shows that the GR is assigned to team $t(j) = t'_{\bmod (i, n/m)}$.

D. Stage-4

After determining T', in Stage-4, the ARs create and send msg to the leaf GRs in $\mathcal{T}$ so that the *message relaying* step can start (refer to Algorithm 2, lines 1-6). Our goal is to determine a set of tours for the ARs such that each leaf GR is visited exactly once by only one AR and the maximum tour length among the ARs is minimized. Therefore, we treat the problem as the Multiple Traveling Salesman Problem (MTSP) [32] which is a generalization of the well-known Traveling Salesman Problem (TSP). We use a genetic algorithm to solve this problem. The MTSP solver takes the locations of the leaf GRs, number of ARs, minimum number of GRs to be visited by each AR, size of the population, and number of iterations as input, and returns the tour for each AR (refer to Algorithm 1, line 34-35). Since MTSP solver is executed when the exploration stage is completed, each AR's tour starts and ends at $\mathcal{X}_{r^*}$.

E. Stage-5

For a relay GR, the message relaying phase finishes after the GR relays the msg to its parent, whereas for a non-relay GR, this phase finishes as soon as the GR receives the msg. In Stage-5, we determine the final configurations of the GRs and AR in a team such that the team forms a fully connected network topology.

Let $c(t_i')$ be the center of the circle with the smallest possible radius that encloses all GRs in team t_i'. The radius of this circle is denoted by $c_r(t_i')$ and equal to the furthest pairwise distance among the GRs in t_i'. This problem is known as the smallest-circle problem [33]. Using the homothety transformation in (1), we compute the final configuration of the GR in t_i' as

$$x^f = c(t_i') + \min(1, R/c_r(t_i'))(x' - c(t_i')), \tag{3}$$

where x and x' denote current and the final positions of the GR, respectively. Applying (3) with $\phi = \min(1, R/c_r(t_i'))$ for each GR ensures that the distance between the furthest pair GRs in the team is at most R. Finally, the GR moves to x^f (refer to Algorithm 2, lines 7-9).

The assignment of a single AR to each team is done according to the distance traveled costs by the ARs until the end of Stage-4. Let $tempT$ and $tempU$ keep T' and the set of ARs, respectively. First, $tempU$ is sorted in descending order according to the distance traveled costs by the ARs. The sorted list is denoted by $sortedU$. Until $sortedU$ is empty, we remove the top AR denoted by $top(sortedU)$, assign it to the team t' in $tempT$ whose $c(t')$ is closest to $top(sortedU)$, and remove t' from $tempT$. Finally, AR $top(sortedU)$ moves to $c(t_i')$ (refer to Algorithm 1, lines 44-53).

F. Example Execution

Fig. 1 and Fig. 2 show an example execution from the implementation of CTB algorithm in MATLAB. $n = 24$ GRs are distributed uniformly at random in a circular area of radius $L = 50$. $m = 4$ ARs are initally located at (0,0). $R = 4$ and $h = 2$. Fig. 1 shows the exploration of the ARs in Stage-1. Fig. 2:A-E show the built MST, initial teams (T), final teams (T'), paths of the ARs after executing MTSP Solver, relocation of the GRs in the message relaying step, and final relocation of the GRs so that there is a direct communication link between all pair of robots within their teams, respectively.

IV. SIMULATIONS

In this section, we present the results of the simulations that are conducted to evaluate the performance of algorithm CTB presented in Section III. CTB is implemented in MATLAB. The robots are distributed uniformly at random in a circular area of radius L. L is varied between 100 and 200, n (number

Algorithm 1: Connected Team Builder - AR

Input : L: environment radius, m: number of ARs, R: communication range, h: altitude

1 $u \leftarrow id$ of the AR
`// === Stage-2 ===`
2 $G(V,E) \leftarrow$ Complete graph on X
3 [$\mathcal{T}$] = MST(G)
4 $S^* \leftarrow$ Set of all GRs
5 **foreach** $s \in S^*$ **do**
6 $s.parent = \mathcal{T}.Parents(s)$
7 $x_s \leftarrow$ Position of s
8 **if** *s is the closest child to $s.parent$* **then**
9 $c \leftarrow$ Position of $s.parent$
10 $l = \|c - x_s\|$
11 $x_s' = c + \min(1, R/l)(x_s - c)$
12 $X' = X' \cup x_s'$
13 **else**
14 $X' = X' \cup x_s$
15 **end**
16 **end**
`// === Stage-3 ===`
17 [T, $Centers$] = KMEANSpp(X', m)
18 $M \leftarrow n \times n$ matrix
19 $D \leftarrow$ Array of size n
20 **foreach** *row* $i \in M$ **do**
21 **foreach** *column* $j \in M$ **do**
22 Temporarily remove GR-j from its team $t(j)$
23 Add it to team t_i
24 $index = 0$
25 **foreach** $s \in S^*$ **do**
26 $D[index] = \|x_s' - Centers(\text{t(s)})\|$
27 $index = index + 1$
28 **end**
29 $SE = \sigma(D)/n$
30 $M_{i,j} = SE$
31 **end**
32 **end**
33 [T']=HA(M)
`// === Stage-4 ===`
34 $X(\mathcal{T}.LeafNodes) \leftarrow$ Locations of the leaf GRs in $\mathcal{T}$
35 Tours = MTSPsolver($X(\mathcal{T}.LeafNodes)$, m, $mintour$, $popsize$, $numiter$)
36 **foreach** *GR* $s \in Tours(u)$ **do**
37 Move toward s until $\|x(s) - x(u)\| \leq R$
38 $msg \leftarrow$ CreateMessage(X', $\mathcal{T}$, T')
39 Send msg to s
40 **end**
`// === Stage-5 ===`
41 $tempU \leftarrow$ the list of all ARs
42 $tempT \leftarrow T'$
43 $sortedU$ =Sort($tempU$, *criteria*: dec, *comparator*: DistanceTraveled)
44 **while** $\neg$ *isEmpty($sortedU$)* **do**
45 $t' \leftarrow$ The team in $tempT$ whose $c(t')$ is closest to ut
`// c(t') ← Center of the smallest circle enclosing t'`
46 **if** $top(sortedU)$ == u **then**
47 $AssignedTeam = t'$
48 $AssignedTeamCenter = c(t')$
49 **end**
50 Remove $top(sortedU)$ from $sortedU$
51 Remove t' from $tempT$
52 **end**
53 Move to $c(t')$

Algorithm 2: Connected Team Builder - GR

Input : X': Goal positions of the GRs, $\mathcal{T}$: MST, T': Teams

1 $s \leftarrow id$ of the GR
2 $parent = \mathcal{T}.Parents(s)$
3 **if** *closest to the $parent$* **then**
4 Move to $x_s' \in X'$
5 Relay msg to $parent$
6 **end**
7 $c(t_i') \leftarrow$ Center of the smallest circle enclosing t_i' where $s \in t_i'$
8 $x_s^f = c(t_i') + \min(1, R/c_r(t_i'))(x_s' - c(t_i'))$
9 Move to x_s^f

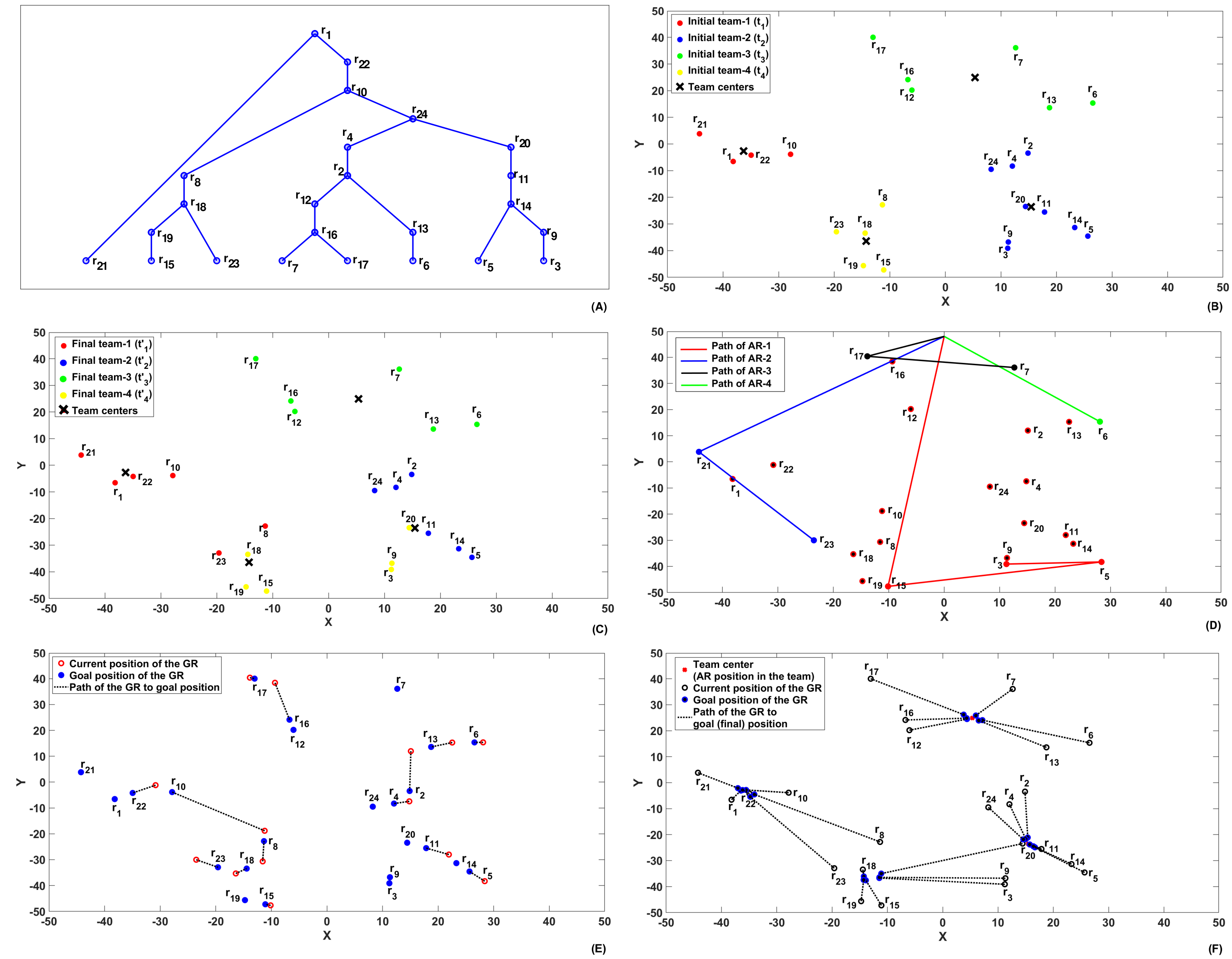

Fig. 2: Example execution of Stage-1 of CTB Algorithm. $L = 50$, $m = 4$, $n = 24$, $R = 4$, and $h = 2$.

of GRs) is varied between 16 and 64, m (number of ARs) is varied between 4 and 32, and R (communication range) is varied between 5 and 40. The result for each tested pair of parameters is averaged over 100 trials. From each trial, we used maximum distance traveled among ARs and the maximum distance traveled among GRs. The speed of each aerial and ground robot is equal to one unit. Thus, time and distance are equivalent. As the MTSP solver, we use the function `mtsp_ga_minmax` implemented in MATLAB [1].

Distance traveled is used as the evaluation metric. The metric is evaluated (1) with respect to the change in L and n for fixed R and m, (2) with respect to the change in L and R for fixed n and m, (3) with respect to the change in m and L for fixed n and R, and (4) with respect to the change in m and R for fixed n and L. The distance traveled by the AR and the GR is presented in Fig. 3:A-D and Fig. 3:E-H, respectively. From the results, we observe that the performance of CTB quantify as a function of L, m, and R. The results where L is varied show that the distance traveled increases as L increases and decreases as R increases. As the number of ARs (m) increases, the GRs at unknown locations are found sooner, thus the distance traveled by the AR decreases. This outcome is shown in Fig. 3:C-D. Note that m also denotes the number of teams that are formed. Therefore, from Fig. 3:G-H, we observe that as the number teams increases, the distance traveled by the GR decreases.

[1] We use the function `mtsp_ga_minmax` implemented by Joseph Kirk in MATLAB.

V. Conclusion

We studied the aerial-ground multi-robot connected team formation problem in an unbounded area. The objective is the robots with limited communication ranges to autonomously establish teams such that each team consists of equal number of ground robots and exactly one aerial robot. Additionally, the final configurations of the robots in each team should form a fully connected network topology. For this problem, we proposed an algorithm that combines distributed decision-

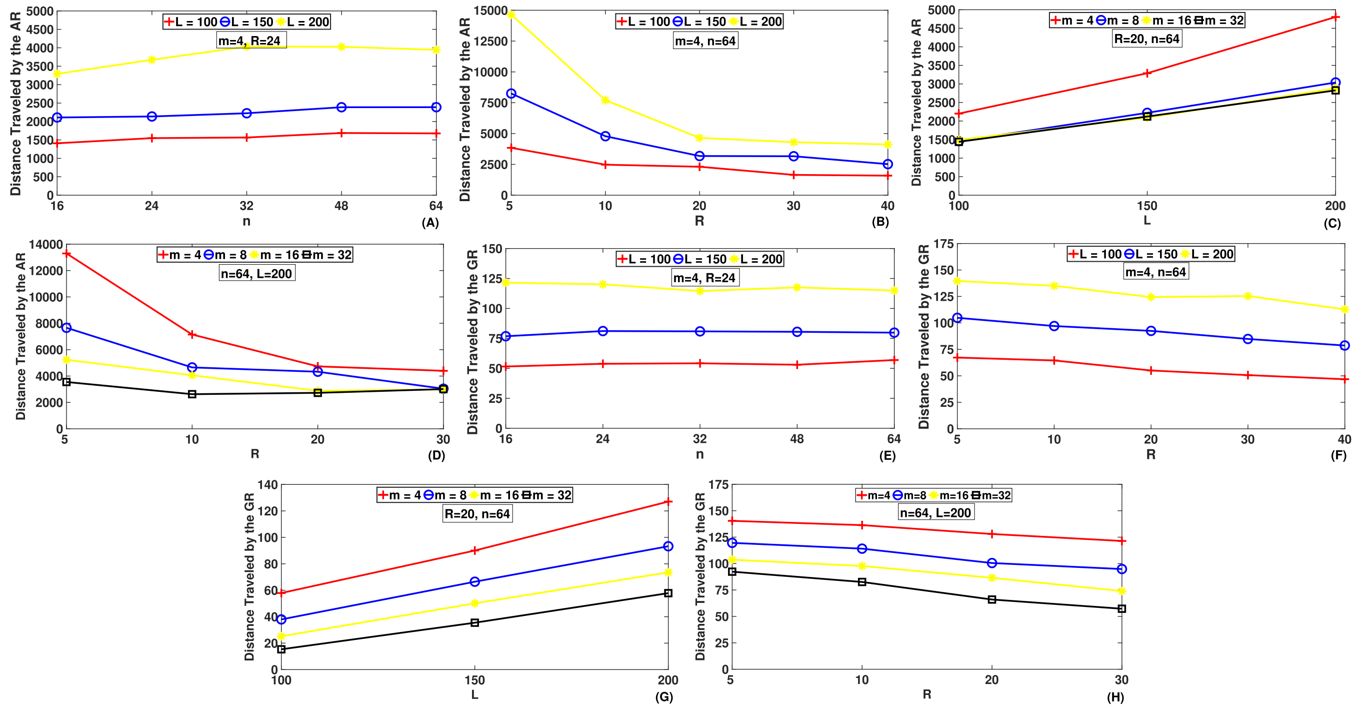

Fig. 3: Simulation results of CTB algorithm. Distance traveled by the **A-D:** aerial robot (AR) and **E-H:** ground robot (GR).

making and online planning, and evaluated its performance in simulations. Our future work includes studying the problem in environments with obstacles.

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